

TMFVC Assignment 11

Guangxin Zhao st194136

January 18, 2025

Exercise 11.1

a)



Code:

```
1 # Exercise 11.1
2
3 # a) Read and inspect the data
4 data <- read.csv("DatasaurusDozen.csv")
5
6 str(data)
7 print(unique(data$dataset))
```

Output:

```
1 'data.frame':  1846 obs. of  3 variables:
2 $ dataset: chr  "dino" "dino" "dino" "dino" ...
3 $ x      : num  55.4 51.5 46.2 42.8 40.8 ...
4 $ y      : num  97.2 96 94.5 91.4 88.3 ...
5 [1] "dino"      "away"      "h_lines"   "v_lines"   "x_shape"
6 [6] "star"      "high_lines" "dots"      "circle"    "bullseye"
7 [11] "slant_up"   "slant_down" "wide_lines"
```

b)

Code:

```
1 # b) Compute the mean and standard deviation of the x and y values for each
  ↪ dataset
2 library(dplyr)
3
4 result <- data %>%
5   group_by(dataset) %>%
6   summarise(
7     mean_x = mean(x),
8     sd_x = sd(x),
9     mean_y = mean(y),
10    sd_y = sd(y))
11
12 print(result)
```

Output:

```
1 # A tibble: 13 × 5
2   dataset    mean_x sd_x mean_y sd_y
3   <chr>      <dbl> <dbl> <dbl> <dbl>
4 1 away      54.3  16.8  47.8  26.9
5 2 bullseye   54.3  16.8  47.8  26.9
6 3 circle     54.3  16.8  47.8  26.9
7 4 dino       54.3  16.8  47.8  26.9
8 5 dots       54.3  16.8  47.8  26.9
9 6 h_lines    54.3  16.8  47.8  26.9
10 7 high_lines 54.3  16.8  47.8  26.9
11 8 slant_down 54.3  16.8  47.8  26.9
12 9 slant_up   54.3  16.8  47.8  26.9
13 10 star      54.3  16.8  47.8  26.9
14 11 v_lines   54.3  16.8  47.8  26.9
15 12 wide_lines 54.3  16.8  47.8  26.9
16 13 x_shape   54.3  16.8  47.8  26.9
```

c)

Code:

```
1 # c) Scatter plot of the data
2 library(ggplot2)
3
4 data %>%
5   ggplot(aes(x = x, y = y, group = dataset)) +
6   geom_point() +
7   facet_wrap(~dataset)
```

Output:

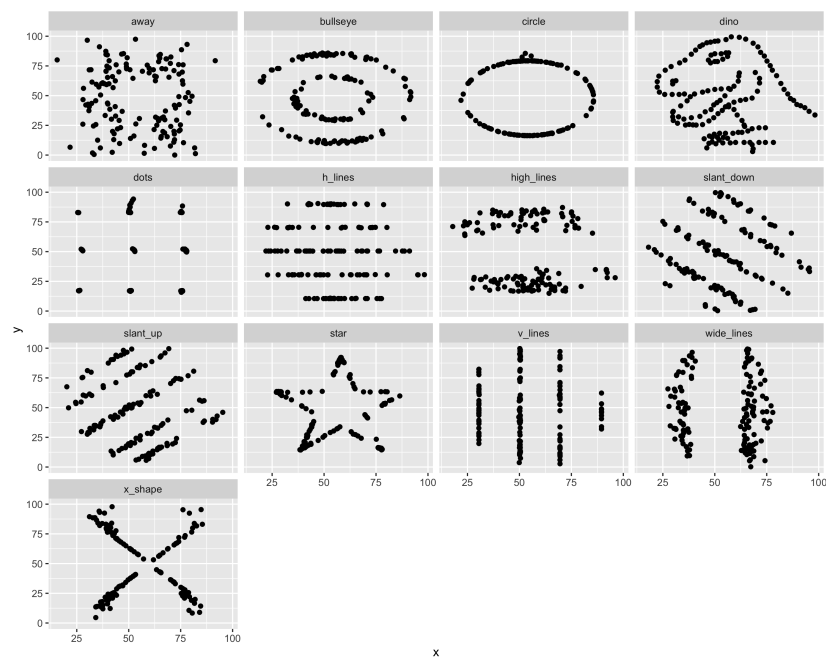


Figure 1: Scatter Plot of the Data

d)

The mean and standard deviation for the x and y variables for each group in the "dataset" variable are the same. However, as shown in Fig. 1, the datasets have different shapes and patterns. This demonstrates that summary statistics alone are insufficient to understand data distributions. Visualization is essential to better identify underlying structures.

Exercise 11.2



a)

1. The result is statistically significant.
2. We can reject H_0 .



b)

Effect size gives an objective, ~~standardized measure~~ of the magnitude of the observed effect.

Two common measures for effect sizes are **Cohen's d** and **Pearson's r**.



c)

Two things one can do with power analysis is **a priori power analysis** and ~~sensitivity analysis~~.



d)

- **Benefits:**

1. ~~Reduces the number of errors in the data.~~
2. Improves the accuracy of statistical metrics such as mean, variance, and correlation.
3. ~~Reduces the risk of overfitting by allowing models to focus on the majority of data points.~~

- **Drawbacks:**

1. ~~Leads to subjective decisions~~ when deleting valuable information.
2. May introduce bias if outliers are removed arbitrarily or without proper justification.
3. Potentially oversimplifies the data by removing valid complexities or underlying patterns.



Exercise 11.3



a)

I would perform **Shapiro-Wilk** since it is more sensitive and powerful to detect a significant effect on small sample sizes.



b)

I can then **visually** check the normality of the data (e.g., using a **histogram** or **Q-Q plot**). I can also apply **data transformation**, such as **log transformation** or **square-root transformation**.

